

# An Approximate Capacity Bound for Sparse Combinatorial Key-Value Memory: Full Derivation and Validation

Supplementary note to “Combinatorial Coding, Key-Value Recall, and Emergent Constraint Satisfaction in a Hippocampal-Inspired Spiking Network”

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## Abstract

This note gives the complete derivation, with every intermediate step made explicit, of the approximate capacity bound used in the main paper (Section 2.3): an expression for the largest number of items  $M$  a sparse  $K$ -of- $H$  combinatorial key-value memory can store while remaining reliably retrievable under a given level of query corruption. The bound is structurally the same as modern Hopfield networks’ proven exponential capacity [1], but is derived here for a discrete, sparse code via elementary combinatorics and extreme value theory rather than for dense continuous patterns via an energy-function argument. We include a full, self-contained proof of the one non-elementary step (the expected maximum of many correlated-in-effect-but-treated-as-independent Gaussian similarities), validate the final bound against simulation, and discuss its relationship to the wider family of capacity theorems it belongs to. We then go further than that first-moment approximation: Section 8 derives an exact, non-asymptotic failure probability directly from the hypergeometric distribution, and a corresponding rigorous sufficient bound on capacity at any chosen target reliability  $\delta$ , correcting a subtle correlation error that a natural-looking marginalization over query corruption would otherwise introduce. This exact bound is, as expected, considerably more conservative than the first-moment approximation, and is shown to be tight in the well-separated regime and honestly, quantifiably loose only as the system approaches its collapse transition.

## 1 Setup

We store  $M$  items. Item  $m$ ’s key is a binary vector  $h_m \in \{0, 1\}^H$  with exactly  $K$  active (value-1) coordinates – a  $K$ -of- $H$  sparse code. Throughout this note we model the  $M$  stored keys as independent, uniformly random  $K$ -subsets of  $[H] = \{1, \dots, H\}$ : an idealization relative to the true architecture, where keys are  $W_{SH}$ -projections of mutually correlated combinatorial structure (solved Sudoku grids), addressed explicitly in Section 9.

A query is formed from the true key  $h_{\text{true}}$  by flipping a fraction  $\eta \in [0, 1]$  of all  $H$  coordinates, chosen uniformly at random without replacement – the noise model used throughout the main paper. Retrieval compares the query against all  $M$  stored keys via similarity (shared active coordinates) and should return the index of  $h_{\text{true}}$ . Retrieval *fails* when some other stored key’s similarity to the query exceeds  $h_{\text{true}}$ ’s.

We ask: for fixed  $H$ ,  $K$ , and corruption level  $\eta$ , how large can  $M$  be before this kind of failure becomes likely?

## 2 Step 1: Similarity between two independent random keys

**Lemma 1** (Hypergeometric overlap). *Let  $A, B \subset [H]$  be independent uniformly random subsets with  $|A| = a$ ,  $|B| = b$ . Then  $|A \cap B|$  is a hypergeometric random variable with*

$$\mathbb{E}[|A \cap B|] = \frac{ab}{H}, \quad \text{Var}(|A \cap B|) = a \cdot \frac{b}{H} \left(1 - \frac{b}{H}\right) \cdot \frac{H - a}{H - 1}.$$

*Proof sketch.* Fix  $A$  and treat  $B$ 's  $b$  elements as drawn sequentially without replacement from  $[H]$ . Let  $X_i = \mathbb{I}[\text{the } i\text{-th draw lands in } A]$ , so  $|A \cap B| = \sum_{i=1}^b X_i$ . Each  $X_i$  is Bernoulli( $a/H$ ) by symmetry, giving the mean by linearity of expectation. For the variance,  $\text{Cov}(X_i, X_j) < 0$  for  $i \neq j$  because sampling without replacement makes draws negatively associated (having drawn a marked element makes drawing another marked element on a later draw slightly less likely); summing the  $b$  variances and  $\binom{b}{2}$  covariances gives the stated finite-population-corrected formula. (Full derivation: Feller, *An Introduction to Probability Theory*, Vol. 1, Ch. II.)  $\square$

Applying Lemma 1 with  $a = b = K$  gives the similarity between two *uncorrupted* stored keys: mean  $K^2/H$ , the number quoted throughout the main paper (e.g.  $K^2/H = 22.5$  for the default  $H = 1000$ ,  $K = 150$ ).

## 3 Step 2: Statistics of a corrupted query

Flipping a uniformly random fraction  $\eta$  of all  $H$  coordinates is itself an instance of Lemma 1: the  $\eta H$  flipped positions are a random subset of  $[H]$ , and we ask how many of  $h_{\text{true}}$ 's  $K$  active positions they contain. By the lemma (with  $a = K$ ,  $b = \eta H$ ), the expected number of active positions flipped *off* is  $\eta K$ , so

$$S_{\text{true}}(\eta) = \mathbb{E}[\text{similarity of query to } h_{\text{true}}] = K(1 - \eta), \quad (1)$$

exactly (no further approximation needed: this is a direct expectation, not a bound). The expected number of originally-*inactive* positions flipped *on* is, symmetrically,  $\eta(H - K)$ , so the corrupted query's own total active-coordinate count is, in expectation,

$$K_q(\eta) = K(1 - \eta) + \eta(H - K) = K(1 - 2\eta) + \eta H. \quad (2)$$

The query's similarity to any *other*, independent stored key is now Lemma 1 applied with  $a = K_q(\eta)$ ,  $b = K$ :

$$\mu_{\text{other}}(\eta) = \frac{K_q(\eta) K}{H}, \quad \sigma_{\text{other}}^2(\eta) = K_q(\eta) \frac{K}{H} \left(1 - \frac{K}{H}\right) \frac{H - K_q(\eta)}{H - 1}. \quad (3)$$

## 4 Step 3: Gaussian approximation

The hypergeometric distribution of  $S_{\text{other}}$  (similarity to one specific other key) is exact but discrete and unwieldy for the extreme-value step that follows. We approximate

$$S_{\text{other}} \approx \mathcal{N}(\mu_{\text{other}}(\eta), \sigma_{\text{other}}^2(\eta)),$$

justified for  $H \gg 1$  by the central limit theorem for sampling without replacement from a finite population [2, 3]: sums arising from simple random sampling without replacement are asymptotically normal as the population size grows, with exactly the finite-population-corrected variance already appearing in Lemma 1. This is the least exact step in the derivation, and its error is largest precisely when  $\sigma_{\text{other}}$  is small relative to the mean – i.e. at low noise, where the discreteness of the underlying hypergeometric distribution matters most. Section 7 quantifies this error directly rather than leaving it as a caveat.

## 5 Step 4: The maximum of many competitors

Retrieval does not fail when the *typical* competing key looks similar to the query – it fails when the *single most similar* of the  $M - 1$  competitors exceeds  $S_{\text{true}}(\eta)$ . This is an extreme-value criterion, and it is the step that most distinguishes this derivation from a naive signal-to-noise argument.

**Lemma 2** (Expected maximum of i.i.d. Gaussians). *Let  $Z_1, \dots, Z_n$  be i.i.d.  $\mathcal{N}(0, 1)$ . Then*

$$\mathbb{E} \left[ \max_{1 \leq i \leq n} Z_i \right] \leq \sqrt{2 \ln n} + o(1) \quad \text{as } n \rightarrow \infty,$$

*and this bound is asymptotically tight:  $\mathbb{E}[\max_i Z_i] = \sqrt{2 \ln n} (1 + o(1))$ .*

*Proof of the upper bound.* For any  $t \geq 0$ , the standard Gaussian tail (Chernoff) bound gives  $\Pr(Z_1 > t) \leq e^{-t^2/2}$ : with  $s = t$ ,  $\Pr(Z_1 > t) = \Pr(e^{sZ_1} > e^{st}) \leq \mathbb{E}[e^{sZ_1}]e^{-st} = e^{s^2/2-st}$ , minimized at  $s = t$ . A union bound over the  $n$  variables gives  $\Pr(\max_i Z_i > t) \leq \min(1, ne^{-t^2/2})$ . Writing the expectation of a nonnegative random variable as an integral of its tail,

$$\mathbb{E} \left[ \max_i Z_i \right] = \int_0^\infty \Pr \left( \max_i Z_i > t \right) dt \leq \int_0^{t^*} 1 dt + \int_{t^*}^\infty ne^{-t^2/2} dt,$$

where  $t^* = \sqrt{2 \ln n}$  is chosen so the two bounds on the integrand meet ( $ne^{-(t^*)^2/2} = 1$ ). The first integral is exactly  $t^* = \sqrt{2 \ln n}$ . For the second, the standard Gaussian tail estimate  $\int_a^\infty e^{-t^2/2} dt \leq \frac{1}{a} e^{-a^2/2}$  (for  $a > 0$ , obtained by writing  $e^{-t^2/2} \leq \frac{t}{a} e^{-t^2/2}$  on  $t \geq a$  and integrating exactly) gives

$$\int_{t^*}^\infty ne^{-t^2/2} dt \leq \frac{n}{t^*} e^{-(t^*)^2/2} = \frac{1}{\sqrt{2 \ln n}} \xrightarrow{n \rightarrow \infty} 0.$$

Summing the two pieces gives  $\mathbb{E}[\max_i Z_i] \leq \sqrt{2 \ln n} + o(1)$ . The matching lower bound requires a second-moment (Paley–Zygmund-type) argument and is omitted here; see Boucheron, Lugosi & Massart, *Concentration Inequalities* [4], Ch. 2, for the complete two-sided result.  $\square$

Rescaling Lemma 2 to  $\mathcal{N}(\mu_{\text{other}}, \sigma_{\text{other}}^2)$  variables and setting  $n = M - 1$  (the number of competing stored keys) gives the working approximation used in the main paper:

$$\mathbb{E} \left[ \max_{i=1}^{M-1} S_{\text{other},i} \right] \approx \mu_{\text{other}}(\eta) + \sigma_{\text{other}}(\eta) \sqrt{2 \ln(M-1)}. \quad (4)$$

## 6 Step 5: Solving for capacity

Retrieval is (approximately, at the level of expectations rather than a proper tail bound on failure probability) reliable when  $S_{\text{true}}(\eta)$  exceeds the expected maximum competitor, Equation 4. Define the normalized separation margin

$$z(\eta) = \frac{S_{\text{true}}(\eta) - \mu_{\text{other}}(\eta)}{\sigma_{\text{other}}(\eta)}. \quad (5)$$

Setting  $S_{\text{true}}(\eta) = \mu_{\text{other}}(\eta) + \sigma_{\text{other}}(\eta) \sqrt{2 \ln(M-1)}$  and solving for  $M$ :

$$z(\eta) = \sqrt{2 \ln(M-1)} \iff M-1 = \exp\left(\frac{z(\eta)^2}{2}\right) \iff \boxed{M_{\text{max}}(\eta) \approx 1 + \exp\left(\frac{z(\eta)^2}{2}\right)}. \quad (6)$$

**Proposition 1** (Structural form). *Equation 6 states that capacity is exponential in the square of a normalized signal-to-noise-like quantity. This is the same structural form as modern Hopfield networks’ proven capacity  $C \cong 2^{d/2}$  [1], though that bound is derived via a Lyapunov energy-function argument (the Concave–Convex Procedure applied to a log-sum-exp energy) over dense continuous patterns, not via hypergeometric statistics over a sparse discrete code.*

## 7 Validation against simulation

Equations 1–6 were implemented exactly as derived (`capacity_bound_derivation.py`, no free parameters fit to data) and compared against the empirically measured collapse thresholds reported in the main paper.

|                                  | $M=50$ | $M=1000$ | $M=10,000$ | $K=10$ | $K=150$ |
|----------------------------------|--------|----------|------------|--------|---------|
| Predicted (this note, $H=1000$ ) | 43.9%  | 41.9%    | 40.6%      | 24.5%  | 41.9%   |
| Observed (simulation)            | 45%    | 45%      | 40%        | 25%    | 40–45%  |

Agreement is within 1–5 percentage points across a 200-fold range of  $M$  and an order of magnitude in  $K$ , including an almost exact match at  $K = 10$ . A second, independent check: at  $H=1000$ ,  $K=150$ ,  $M=200$ , the derivation predicts the ratio  $S_{\text{true}}(\eta)/\mathbb{E}[\max_i S_{\text{other},i}]$  at  $\eta \in \{0, 0.20, 0.30\}$  as 4.21, 1.99, 1.47 respectively, against independently measured ratios of 3.39, 1.92, 1.45 (obtained from a completely separate experiment, the separation-operator ablation, not used to fit any parameter of this derivation). The two noisy points match to within 4% and 1.4%; the noise-free point is over-predicted by about 24%, consistent with Section 4’s observation that the Gaussian approximation is weakest exactly there.

## 8 An exact hypergeometric tail bound

Equation 6 matches only the *expected* maximum competitor similarity, not a probabilistic guarantee on retrieval failure – a gap noted below in Section 9. This section closes it with an exact, non-asymptotic tail bound on the failure probability itself, derived directly from the hypergeometric distribution rather than its Gaussian approximation, and valid at any finite  $M$ .

### 8.1 A correlation a naive marginalization misses

Let  $R$  denote the (random) number of  $h_{\text{true}}$ ’s  $K$  active positions that land among the  $\eta H$  flipped query positions. By Lemma 1 (with  $a = K$ ,  $b = \eta H$ ),  $R$  is exactly hypergeometric:

$$R \sim \text{Hypergeometric}(H, K, \eta H), \quad \Pr(R = r) = \frac{\binom{K}{r} \binom{H-K}{\eta H - r}}{\binom{H}{\eta H}}.$$

Conditional on  $R = r$ , both the query’s active-coordinate count and its similarity to  $h_{\text{true}}$  are *deterministic*, not merely expected:

$$k(r) = K + \eta H - 2r, \quad s(r) = K - r,$$

the realized versions of Equations 2 and 1, which recover  $k(\eta K) = K_q(\eta)$  and  $s(\eta K) = S_{\text{true}}(\eta)$  only in expectation. Crucially,  $k(r)$  and  $s(r)$  are perfectly correlated through the same draw of  $r$ :

a query realization with more of its true bits flipped off (large  $r$ ) simultaneously has lower true-similarity *and* a different active count. A natural-looking but invalid shortcut is to hold  $s(r)$  fixed at its unconditional expectation  $S_{\text{true}}(\eta) = K(1 - \eta)$  while still letting  $k(r)$  vary with  $r$  inside the union bound over competitors: since  $k(r)$  and  $s(r)$  are deterministic functions of the very same random variable, fixing one while randomizing the other breaks their correlation and is not a valid marginalization of any well-defined random process. Numerically, this shortcut understates the true failure probability by roughly four to five orders of magnitude for  $\eta \in [0.30, 0.40]$ , converging to the correct value only as  $\eta$  approaches the collapse region (where  $r$ 's distribution concentrates and the distinction between fixing  $s(r)$  at  $r$  or at  $\mathbb{E}[r]$  stops mattering).

## 8.2 Exact per-realization failure probability

Fix one specific other stored key  $h_j$  ( $j \neq \text{true}$ ), itself an independent random  $K$ -subset of  $[H]$ . Conditional on  $R = r$ , the query is a specific, realized  $k(r)$ -subset, and its overlap with  $h_j$  is exactly hypergeometric by Lemma 1 (with  $a = k(r)$ ,  $b = K$ ). Define the exact tail probability

$$p_k(s) := \Pr [\text{Hypergeometric}(H, k, K) \geq s] = \sum_{i=s}^{\min(k, K)} \frac{\binom{k}{i} \binom{H-k}{K-i}}{\binom{H}{K}}.$$

(We use  $\geq$  rather than strict  $>$ : an exact tie is conservatively counted as a failure, which can only inflate the resulting probability and keeps Proposition 2 below a genuine upper bound regardless of how ties are broken at retrieval time.) Conditional on  $R = r$ , the probability that this one specific competitor's similarity meets or exceeds  $h_{\text{true}}$ 's is exactly  $p_{k(r)}(s(r))$ , and with  $M - 1$  independent competitors the conditional failure probability is exactly

$$\Pr[\text{fail} \mid R = r] = 1 - [1 - p_{k(r)}(s(r))]^{M-1}.$$

Averaging over  $R$  gives the exact unconditional failure probability – no Gaussian approximation, no expectation-of-a-maximum heuristic, and no fixing of  $s(r)$  at its mean:

$$P_{\text{fail}}(M, \eta) = \sum_{r=0}^K \Pr(R = r) \cdot \left\{ 1 - [1 - p_{k(r)}(s(r))]^{M-1} \right\}. \quad (7)$$

Equation 7 is exact given the independence assumption of Section 9; it involves only a finite sum over  $r \in \{0, \dots, K\}$  and is cheap to evaluate for any  $H, K, \eta, M$ .

## 8.3 No finite-size singularity: monotonicity and convexity in $M$

Equation 7 is not just numerically well-behaved (Section 8.5 below) – it is provably free of any finite- $M$  singularity of the kind a genuine attractor-collapse memory cliff would produce. This is a structural property of the exact formula itself, independent of the union bound derived in the next section, and is worth stating as a theorem in its own right.

**Theorem 1** (No finite-size singularity). *Fix  $H$ ,  $K$ , and  $\eta$ , and suppose  $0 < p_{k(r)}(s(r)) < 1$  for every  $r$  with  $\Pr(R = r) > 0$  (against any realized corruption level, a single competitor's failure is neither impossible nor certain). Extend Equation 7 to continuous  $M \geq 1$  via*

$$P_{\text{ret}}(M, \eta) := 1 - P_{\text{fail}}(M, \eta) = \sum_{r=0}^K \Pr(R = r) [1 - p_{k(r)}(s(r))]^{M-1}.$$

Then  $P_{\text{ret}}(M, \eta)$  is strictly decreasing and strictly convex in  $M$ :

$$\frac{\partial P_{\text{ret}}}{\partial M} < 0, \quad \frac{\partial^2 P_{\text{ret}}}{\partial M^2} > 0.$$

*Proof.* Write  $q_r := 1 - p_{k(r)}(s(r)) \in (0, 1)$  for each  $r$  with  $\Pr(R = r) > 0$ . Treating  $M$  as a continuous variable,  $q_r^{M-1} = \exp[(M-1) \ln q_r]$ , so

$$\frac{\partial}{\partial M} q_r^{M-1} = q_r^{M-1} \ln q_r, \quad \frac{\partial^2}{\partial M^2} q_r^{M-1} = q_r^{M-1} (\ln q_r)^2.$$

Since  $q_r \in (0, 1)$ ,  $\ln q_r < 0$  and  $q_r^{M-1} > 0$ , so the first derivative is strictly negative and the second, a square of a nonzero real number times a positive factor, is strictly positive.  $P_{\text{ret}}(M, \eta)$  is a finite sum of these terms weighted by  $\Pr(R = r) \geq 0$  summing to 1; differentiation is linear and commutes with a finite sum, so

$$\frac{\partial P_{\text{ret}}}{\partial M} = \sum_r \Pr(R = r) q_r^{M-1} \ln q_r < 0, \quad \frac{\partial^2 P_{\text{ret}}}{\partial M^2} = \sum_r \Pr(R = r) q_r^{M-1} (\ln q_r)^2 > 0,$$

each a weighted average, with at least one strictly positive weight, of terms that are all strictly negative (respectively all strictly positive).  $\square$

**Interpretation.** Theorem 1 rules out, at the level of the exact model, any finite- $M$  point at which retrieval probability could drop discontinuously or bend toward an accelerating collapse:  $P_{\text{ret}}$  can only *decelerate* as  $M$  grows (convexity), on top of always decreasing (monotonicity) – graceful, non-catastrophic degradation, now proven rather than only observed in simulation (Section 8.5). This is a property specific to the *explicit-key* retrieval mechanism analyzed throughout this note: failure accumulates from more chances for some one unrelated stored key to accidentally resemble the query, not from the appearance or disappearance of a dynamical attractor – the latter being the mechanism behind a genuine finite-size memory cliff in attractor-based (Hopfield-type) associative memories, and a distinction this note does not claim to resolve for that broader model class (Section 9).

#### 8.4 A rigorous, closed-form sufficient bound

Equation 7 is exact but implicit: it gives  $P_{\text{fail}}$  as a function of  $M$ , not  $M$  as a function of a target failure probability  $\delta$ . We invert it via the union bound, applied *per realization of  $r$*  before averaging – the order matters, and doing it in this order is what keeps the resulting bound valid.

By Bernoulli's inequality,  $1 - (1 - p)^n \leq np$  for any  $p \in [0, 1]$  and integer  $n \geq 0$ . Applied pointwise, for every  $r$ ,

$$1 - [1 - p_{k(r)}(s(r))]^{M-1} \leq (M-1) p_{k(r)}(s(r)).$$

This is a deterministic inequality between two functions of  $r$ ; taking  $\mathbb{E}_R[\cdot]$  of both sides preserves it, since expectation is monotone:

$$P_{\text{fail}}(M, \eta) \leq (M-1) \cdot \underbrace{\mathbb{E}_R[p_{k(R)}(s(R))]}_{=: \bar{P}(\eta)}. \quad (8)$$

**Proposition 2** (Corrected sufficient capacity bound). *For any target failure probability  $\delta \in (0, 1)$ , storing*

$$M \leq M_{\max}(\eta, \delta) := 1 + \left\lfloor \frac{\delta}{\bar{P}(\eta)} \right\rfloor \quad (9)$$

*items guarantees  $P_{\text{fail}}(M, \eta) \leq \delta$  exactly, subject only to the independence assumption of Section 9 – not to any Gaussian or large- $M$  asymptotic approximation.*

*Proof.* Immediate from Equation 8:  $(M - 1)\bar{P}(\eta) \leq \delta$  whenever  $M - 1 \leq \delta/\bar{P}(\eta)$ , i.e.  $M \leq 1 + \lfloor \delta/\bar{P}(\eta) \rfloor$ .  $\square$

Equation 9 has the same union-bound shape as a natural first attempt at this formula,  $1 + \lfloor \delta/p_{K_q(\eta)}(S_{\text{true}}(\eta)) \rfloor$ , but with  $p$  correctly taken to be  $\bar{P}(\eta) = \mathbb{E}_R[p_{k(R)}(s(R))]$  – the per-realization tail probability *properly averaged* over  $R$  – rather than a single tail probability evaluated at the mismatched pair  $(K_q(\eta), S_{\text{true}}(\eta))$ , which (Section 8.1) treats the numerator and denominator of the same random draw as independently settable.

## 8.5 Numerical validation

Table 1 reports  $\bar{P}(\eta)$  and the resulting  $M_{\max}(\eta, \delta)$  at  $H = 1000$ ,  $K = 150$  for two target failure probabilities, alongside  $M_{\max}(\eta)$  from Equation 6 (Section 6) for comparison. All quantities were computed with `scipy.stats.hypergeom` and, where probabilities are extremely small, the numerically stable form  $1 - (1 - p)^n = -\text{expm1}(n \cdot \log1p(-p))$  to avoid catastrophic cancellation.

| $\eta$ | $\bar{P}(\eta)$        | $M_{\max}(\eta, \delta=0.01)$ | $M_{\max}(\eta, \delta=0.05)$ | $M_{\max}(\eta)$ [Eq. 6] |
|--------|------------------------|-------------------------------|-------------------------------|--------------------------|
| 0.20   | $2.41 \times 10^{-35}$ | $4.15 \times 10^{32}$         | $2.07 \times 10^{33}$         | $2.29 \times 10^{48}$    |
| 0.30   | $8.26 \times 10^{-15}$ | $1.21 \times 10^{12}$         | $6.05 \times 10^{12}$         | $1.61 \times 10^{19}$    |
| 0.35   | $9.59 \times 10^{-9}$  | $1.04 \times 10^6$            | $5.21 \times 10^6$            | $2.61 \times 10^{10}$    |
| 0.40   | $1.22 \times 10^{-4}$  | 83.0                          | 411                           | $3.27 \times 10^4$       |
| 0.42   | $1.81 \times 10^{-3}$  | 6.0                           | 28.0                          | 739                      |
| 0.45   | $3.73 \times 10^{-2}$  | 1.0                           | 2.0                           | 13.9                     |

Table 1: Corrected exact sufficient bound versus the expected-maximum approximation (both at  $H = 1000$ ,  $K = 150$ ).

Two things stand out. First, the exact tail-based bound is dramatically more conservative than the expected-maximum approximation – by about seven orders of magnitude at  $\eta = 0.30$  ( $1.2 \times 10^{12}$  vs.  $1.6 \times 10^{19}$ ), narrowing to about a factor of 120 at  $\eta = 0.42$ . This gap is expected, not a contradiction between the two derivations: Equation 6 only requires the true match to beat the *expected value* of the maximum competitor, a much weaker requirement than controlling the actual *probability* of failure to a small  $\delta$  via a union bound over  $M - 1$  competitors. The two bounds answer different questions – “typical-case margin” versus “guaranteed failure rate” – and are best read together: Equation 6 explains *why* capacity is exponential in the squared margin (the mechanism), while Equation 9 gives an actually-defensible number at a chosen reliability level (the guarantee). Section 9’s prediction that a proper tail bound “would very likely give a somewhat smaller  $M_{\max}$ ” is confirmed, considerably more starkly than that hedge anticipated.

Second, we check how loose the sufficient bound (Equation 9) is relative to the exact failure probability it bounds (Equation 7), by comparing  $(M - 1)\bar{P}(\eta)$  against  $P_{\text{fail}}(M, \eta)$  directly (Table 2).

The bound is essentially tight (ratio  $\approx 1.00$ ) at  $\eta = 0.30$ , and increasingly loose – by design, as any union bound must be once per-competitor probabilities are no longer small – as  $\eta$  approaches

| $\eta$ | $M$  | $P_{\text{fail}}(M, \eta)$ [exact, Eq. 7] | $(M - 1)\bar{P}(\eta)$ [Eq. 8] | ratio |
|--------|------|-------------------------------------------|--------------------------------|-------|
| 0.30   | 200  | $1.644 \times 10^{-12}$                   | $1.644 \times 10^{-12}$        | 1.00  |
| 0.30   | 1000 | $8.252 \times 10^{-12}$                   | $8.252 \times 10^{-12}$        | 1.00  |
| 0.30   | 4000 | $3.303 \times 10^{-11}$                   | $3.303 \times 10^{-11}$        | 1.00  |
| 0.40   | 200  | $1.82 \times 10^{-2}$                     | $2.42 \times 10^{-2}$          | 1.33  |
| 0.40   | 1000 | $5.98 \times 10^{-2}$                     | $1.22 \times 10^{-1}$          | 2.03  |
| 0.40   | 4000 | $1.35 \times 10^{-1}$                     | $4.87 \times 10^{-1}$          | 3.59  |
| 0.42   | 200  | $1.59 \times 10^{-1}$                     | $3.61 \times 10^{-1}$          | 2.27  |
| 0.42   | 1000 | $3.46 \times 10^{-1}$                     | $1.81 \times 10^0$             | 5.23  |
| 0.42   | 4000 | $5.38 \times 10^{-1}$                     | $7.25 \times 10^0$             | 13.46 |

Table 2: Tightness of the sufficient bound against the exact failure probability.

the collapse transition and  $M$  grows (up to a factor of 13.5 at  $\eta = 0.42$ ,  $M = 4000$ ). This is the expected and honest behavior of a union bound: tightest exactly in the well-separated regime where a capacity claim is most meaningful, and openly announcing its own growing conservatism as the system approaches saturation rather than failing silently.

## 9 Limitations

Two assumptions are false by construction and are stated as such rather than absorbed silently.

**Independence.** Real stored keys are  $W_{SH}$ -projections of genuinely correlated combinatorial objects (solved Sudoku grids share row/column/box structure), not independent uniform random subsets. The observed collapse thresholds sit consistently at or slightly above the predicted ones (Section 7); a plausible explanation is that this real correlation makes the worst-case competing key marginally less extreme than the independent model assumes, though this has not been tested directly.

**The extreme-value step is a first-moment approximation, not a tail bound – now resolved.** Equation 4 matches the *expected* maximum, not a high-probability bound on it. Section 8 closes this gap with an exact, non-asymptotic alternative (Equation 7) and a corresponding rigorous sufficient bound (Proposition 2), derived directly from the hypergeometric distribution rather than the Gaussian approximation or the expected-maximum heuristic. As anticipated, the rigorous bound gives a substantially smaller  $M_{\text{max}}$  at any fixed target failure probability – by several orders of magnitude away from the collapse region (Table 1) – and is itself provably tight in that same well-separated regime, growing conservatively (rather than silently failing) only as  $\eta$  approaches collapse (Table 2). Equation 6 should now be read as characterizing the right *mechanism* – exponential capacity in the squared separation margin, hence a sharp rather than gradual collapse – while Equation 9 is the bound to use for an actual reliability guarantee at a chosen  $\delta$ . Both bounds still share the independence assumption below, which remains open.

## 10 Relation to the wider literature

This derivation belongs to a family of results sharing one underlying question: does a correct answer stand out against the *extreme* value among exponentially or combinatorially many near-random alternatives? The same skeleton, worked out through different technical routes, underlies Shannon’s channel coding theorem [5], the Random Energy Model in spin-glass theory [6], Gardner’s

perceptron capacity [7], and the classical Amit–Gutfreund–Sompolinsky analysis of Hopfield network capacity [8]. Kanerva’s original capacity analysis for sparse distributed memory [9] is the closest existing derivation in spirit to this one – also a sparse binary code, also analyzed via combinatorial (there, Hamming-ball volume) arguments – though via a different geometric route than the hypergeometric-overlap statistics used here.

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